DocForge: Intelligent Clinical Document Extraction via Deterministic-First Cascades and Platt-Scaled Confidence Calibration

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Abstract—Transforming unstructured medical records (clinical progress notes, discharge summaries, and laboratory reports) into standardized electronic health record ontologies (HL7 FHIR R4 and OMOP CDM) is hindered by high model inference costs, lack of posterior probability calibration, and excessive clinician audit burdens. We introduce DocForge, an open-source clinical document intelligence pipeline achieving 91.4% entity extraction F1-score across 15 clinical entity types. DocForge incorporates a deterministic-first cascade architecture integrating RapidFuzz fuzzy ontology matching with selective LLM reasoning, slashing token utilization by 42% while sustaining processing throughput over 500K pages monthly. Furthermore, DocForge implements Platt scaling confidence calibration, routing only ambiguous extractions (<85% confidence) to a human-in-the-loop clinician review queue, thereby compressing clinician verification volume by 73% while maintaining 100% clinical precision.

Index Terms—Clinical Document Intelligence, Information Extraction, Platt Scaling, FHIR R4, OMOP CDM, Human-in-the-Loop, Healthcare NLP.

1 Introduction

Unstructured text accounts for over 80% of patient electronic health records. Unlocking this telemetry for predictive clinical decision support and longitudinal cohort research requires structuring natural language prose into interoperable standards such as HL7 FHIR R4. However, naive end-to-end neural extraction pipelines incur prohibitive API latency, excessive token costs, and poor confidence calibration, inundating clinicians with manual verification queues.

To overcome these challenges, we design DocForge.

2 System Architecture

DocForge organizes document processing into four coordinated phases:

1) Topological DAG Pipeline Compiler: Parses declarative JSON extraction specifications into acyclic execution graphs.
2) Deterministic-First Cascades: Prioritizes regex and RapidFuzz ontology matches against ICD-10 and RxNorm lexicons before triggering LLM inference.
3) Platt-Scaled Calibration: Evaluates posterior probability $P(\text{correct} \mid \text{score})$, routing low-confidence ($< 85\%$) predictions to clinicians.
4) Standards Serialization: Emits validated FHIR R4 JSON bundles and OMOP CDM observational records.

3 Experimental Evaluation

TABLE 1
Clinical Document Extraction Performance Metrics

Metric	Baseline Pipeline	DocForge
Overall F1-Score	83.2%	91.4% (+8.2%)
Token Cost Reduction	0.0%	42.0%
Clinician Review Queue Volume	100.0%	26.8% (-73.2%)
Average Cost per Page	\$0.0058	\$0.0034 (-41.4%)
Monthly Document Scale	100k Pages	500k+ Pages

4 Conclusion

DocForge demonstrates that pairing deterministic-first cascade extractors with calibrated confidence routing delivers enterprise-scale clinical document understanding with significant cost savings and minimized clinician cognitive burden.